THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): A knife with a sharp edge cuts easily. Reason (R): A sharp edge exerts large pressure because of its small area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A cylinder of weight 120 N stands on a base of radius 10 cm. Taking pi as 3.14, the pressure is about: 1

(a) 1200 Pa **(b)** 12 Pa **(c)** 3822 Pa **(d)** 38220 Pa

3. Two cyclones, A over warm sea and B moving over land, are compared. Which statement is right? 1

(a) B is likely to be stronger **(b)** A is likely to be stronger
(c) both are equal **(d)** neither has wind

4. Assertion (A): Dams are made thicker at the bottom. Reason (R): Water pressure increases with depth. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A box exerts 2000 Pa. If its base area is halved without changing weight, the pressure becomes: 1

(a) 500 Pa **(b)** 2000 Pa **(c)** 4000 Pa **(d)** 1000 Pa

6. A ship at sea is advised to move away from the path of a cyclone because: 1

(a) wind and waves are strongest near the eye wall
(b) ships are heavy
(c) the sea is calm
(d) the eye is cold

7. Assertion (A): A drinking straw can lift liquid into the mouth. Reason (R): Atmospheric pressure pushes the liquid up when pressure in the straw is reduced. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A block of weight 60 N rests on a table. The force exerted by the table on it is: 1

(a) 30 N **(b)** 0 **(c)** 60 N downward **(d)** 60 N upward

9. **Assertion (A):** A sharp knife cuts better than a blunt one. 1
Reason (R): For the same force, a smaller area of contact gives a larger pressure.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The walls of a dam are made thicker at the bottom. 1
Reason (R): Pressure of water increases with depth.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the pressure under a 5 kg bag (weight about 50 N) that rests on a 0.1 m by 0.05 m face. 2

12. State why weather satellites are used for cyclone tracking. 2

13. Why is there a risk of a roof being blown off in a strong storm? 2

14. A girl of weight 450 N wears high heels with a total contact area of 0.0009 square metre. Find the pressure and comment. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A cube of side 10 cm weighs 50 N. Find the pressure on a table. If a second identical cube is stacked on it, find the new pressure on the table. 3

16. Explain why the speed of wind is greatest near the eye wall. 3

17. Explain why helium balloons rise but ordinary air balloons do not. 3

OR

Compare the pressures exerted by 3
(a) a 70 kg person (about 700 N) on both feet (0.04 square metre) and
(b) the same person on skis of total area 0.4 square metre. Why is it easy to walk on snow with skis?

18. Describe how a barometer reading will vary during the passing of a cyclone and why. 3

OR

What are the harmful effects of storms on agriculture and how may farmers be helped? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Mohan weighs 600 N. When he stands on both feet, each foot has an area of 0.01 square metre in contact with the ground. When he lifts one foot, he stands on one foot only. 5
(i) State the formula for pressure.
(ii) Find the pressure when he stands on both feet.
(iii) Find the pressure when he stands on one foot.
(iv) A camel's wide feet do not sink in the desert sand, but a person's narrow heels do. Explain.

20. A man weighs 800 N and a cow's hoof area is 0.01 square metre for a 4000 N cow with four hooves. Compare the pressure beneath the cow's hooves with the man's pressure when standing on two feet of total area 0.04 square metre. 5

OR

Explain how air pressure and temperature differences between the equator and the poles drive winds. Why do winds curve? 5

END OF QUESTION PAPER — **Total: 40 Marks**